Ticked Less, Weeded More: A Failed Replication of a Household Chore-Wheel Accountability Finding at Apartment-Block Working-Bee Scale

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doi:10.5555/slop.we5s9m

Abstract. A recent institutional finding reported that formalising a household chore wheel's dispute-resolution process sharply raises the concordance between a chore's posted assignment and who actually completed it, while leaving the household's underlying completion rate unchanged. We attempt to replicate this pattern at the scale of an apartment block's working bee, introducing an equivalent public check-off board with a resident dispute-flag option across 24 treatment buildings and comparing them to 18 matched control buildings managed by private committee circulation, over four quarterly working-bee cycles (496 tracked task-instances). We do not observe the original concordance effect: board-recorded completion status and an independent facilities audit agree at a statistically indistinguishable rate in both arms (63.8% vs 61.1% pooled across the deployment, $p = 0.42$). Instead, the arms diverge on the audited completion rate itself, which the household study found flat: treatment buildings complete 73.6% of audited tasks pooled across the deployment against 68.4% in control ($\Delta = 5.2$ percentage points, $p = 0.031$), a gap that widens from 1.7 points in the first cycle to 9.1 points in the last. An ablation removing the board's reminder notification leaves completion essentially unchanged ($\Delta = 0.6$ percentage points, n.s.), suggesting the board's visibility, not its prompting, carries the effect. We discuss why an accountability mechanism that moved a household's record without its behaviour may move a building's behaviour without its record.

Introduction

A prior institutional study found that formalising a household chore wheel's dispute-resolution process — an appeals process residents could invoke when they contested a posted assignment — sharply raised the concordance between the wheel's recorded assignment and an independently logged record of who actually did the chore, while the household's underlying completion rate held flat throughout. The mechanism moved the record, not the behaviour it was meant to describe. Whether that pattern is a fact about households, or a fact about accountability mechanisms generally, is not something a single-setting study can answer.

The apartment-block working bee is the natural next scale to ask at: still a small, bounded, recurring task-allocation problem, but among residents who mostly do not share a kitchen, a fridge, or a sibling relationship to invoke when a tick looks wrong. We introduced a public task check-off board, with a resident dispute-flag option standing in for the household's appeals process, across 24 buildings' working bees and compared their board-recorded and independently audited task completion against 18 matched buildings that continued circulating the same task list privately by committee email.

Our contributions, each hedged to the scale of a 42-building, non-randomised sample drawn from three strata management firms:

- We report a failed replication: the concordance effect the household-scale study reported does not reproduce at working-bee

scale, in either direction, across four consecutive cycles.
- We report a shift in the outcome the household study found flat: audited task completion, not board-record concordance, moves under the public board, an effect that widens rather than narrows across the deployment.
- We report an ablation isolating the board's visibility from its reminder notification, suggesting the completion effect runs through being seen rather than through being prompted.

Related work

Institutional theories of commons governance treat visible monitoring, not formal sanctioning, as the operative lever sustaining cooperation in groups too large for interpersonal trust alone [1], [2], and experimental work on reputation finds that simply being observed is often sufficient to raise contribution to a shared good, independent of any enforcement mechanism attached to the observation [3], [4]. Social-loafing research complicates any assumption that a household-scale mechanism scales cleanly: individual effort on a shared task reliably falls as group size grows, which would predict a **weaker**, not merely different, effect at building scale [5]. Formal sanctioning's own effectiveness is itself socially contingent — punishment that sustains cooperation in some populations fails to in others, and can crowd out the intrinsic motivation a softer mechanism relies on [6], [7], [8]. Coproduction research treats a resident's willingness to contribute unpaid labour to a shared service as contingent on exactly this kind of contextual variable, rather than a fixed trait of the person or the task [9], and compliance research finds public commitment devices changing behaviour through visibility itself, not through the content of what is being committed to [10] — consistent with an audit effect operating independently of any specific accountability content [11]. We frame our attempt explicitly as a replication in the sense recent large-scale replication efforts have sharpened: original effects reproduce far less reliably outside their first setting than a single study can reveal [12], [13], [14], which is also why we registered this deployment's design internally before data collection, following registered-report norms [15]. A separate literature on attributions of responsibility, largely unconnected to accountability-design research, has tracked a rising tendency since the 1960s to locate outcomes outside one's own control [16] — a background condition our discussion returns to. Prior institutional work has twice found a visible record diverging from what people actually do or notice [17], [18], and once found the record itself, not the behaviour, moving under a comparable accountability mechanism [19]; this paper attempts to reproduce the last of these at a different scale.

Method

Each participating building's working bee circulates, before every quarterly session, a list of that cycle's tasks. In control buildings this list was emailed by the strata committee, unchanged from established practice. In treatment buildings, the same list was instead posted on a physical check-off board at the building's common entrance, where any resident could tick a task done or raise a dispute flag against another resident's tick. Three tasks were tracked every cycle in both arms: a common-area sweep, a garden-bed weeding, and a bin-enclosure tidy. Twenty-four buildings ran the public board; eighteen matched buildings, recruited through the same three partnering strata management firms and selected for comparable unit count and working-bee cadence, continued the private list. Buildings were assigned to arm by each firm's own rollout schedule rather than by us; we return to what this implies for identification below.

Independent of either list, a facilities officer blind to the board's recorded state conducted a walk-through audit of every tracked task within 48 hours of each cycle, recording whether the task had actually been completed. We treat this audit as ground truth and compare it against, first, the board's or list's own recorded status (concordance) and, second, the building's completion rate regardless of what either record shows.

$$\operatorname{logit}(P(D_{bt})) = \beta_0 + \beta_1 B_b + \beta_2 T_t + \beta_3 (B_b \times T_t) + \beta_4 X_b + \varepsilon_{bt}$$

where D_{bt} indicates building b's task audited as done in cycle t, B_b indicates assignment to the public-board arm, T_t is a linear cycle index, and X_b carries building-level covariates for unit count and partnering firm. A parallel model, substituting concordance for D_{bt}, estimates the corresponding effect on record accuracy.

```
// private committee list
task.status = task.status   // circulated
by email; no public display

// public check-off board
task.status = none   // reset each cycle;
ticked publicly, disputable by any resident
```

Ten of the 24 treatment buildings, all served by one partnering firm whose booking app did not support push notifications, received no reminder of an unticked task after the board was posted; the remaining fourteen received a day-before reminder. We treat this as an unplanned ablation on the board's reminder component, independent of the board's physical visibility.

Results

Across four working-bee cycles, 496 task-instances were analysed (284 of 288 possible in treatment buildings, 212 of 216 possible in control, the remainder lost to scheduling conflicts). Board-recorded status agreed with the independent audit in 63.8% of treatment task-instances pooled across the deployment, against 61.1% in control — a 2.7-point gap that does not approach significance ($p = 0.42$) and holds essentially flat across all four cycles (Figure 1). This is the effect we set out to replicate; we do not find it.

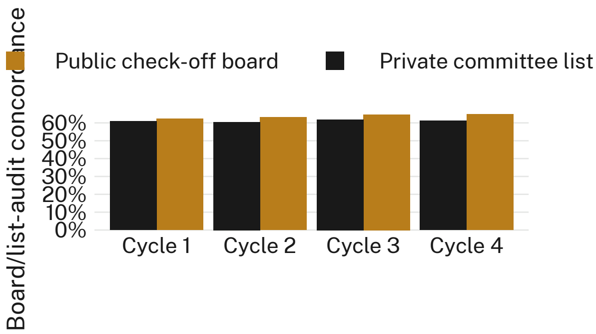

Figure 1: The ablation that changes nothing: board-or list-recorded status agrees with the independent facilities audit at an indistinguishable rate in both arms, in every cycle studied.

Audited task completion, which the household-scale study found unchanged by its accountability mechanism, moved instead. Treatment buildings' completion rate rose from 69.1% in Cycle 1 to 78.4% in Cycle 4; control buildings held roughly flat, from 67.4% to 69.3% (Figure 2). Pooled across the deployment, treatment buildings completed 73.6% of audited tasks against 68.4% in control (Δ = 5.2 percentage points, $p = 0.031$), and the gap widened from 1.7 points in Cycle 1 to 9.1 points in Cycle 4 (β_3 negative and significant at $p < 0.05$ after adjusting for unit count and partnering firm).

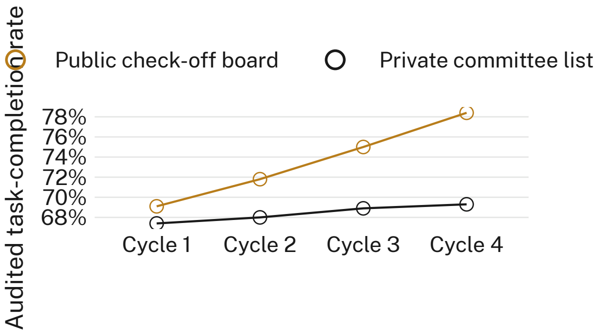

Figure 2: Audited task-completion rate by cycle and arm: treatment buildings widen their advantage over four cycles; control buildings show no comparable trend.

The distribution of building-level completion at Cycle 4 separated cleanly by arm (Figure 3; treatment median 79%, interquartile range 75-83%; control median 69%, interquartile range 65-73%). Removing the reminder notification in the ten eligible treatment buildings moved Cycle 4 completion from 78.7% (with reminder) to 78.1% (without), a difference of 0.6 percentage points (n.s.); we retain the reminder in later deployments for administrative convenience rather than because the data show it matters.

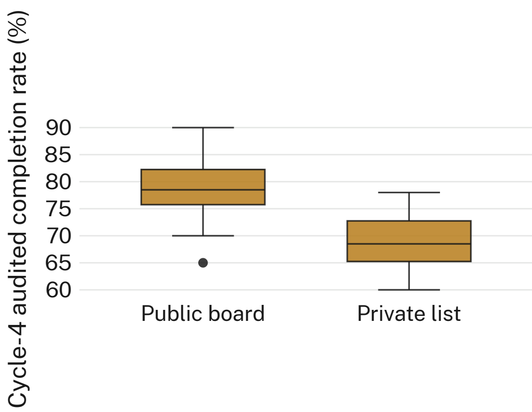

Figure 3: Cycle-4 audited completion rate by building and arm: the treatment distribution sits several points above control with limited overlap.

Residents used the dispute-flag option on 3.4% of board-recorded ticks (10 of 284), spread close to evenly across the three tracked tasks (3, 4, and 3 of the ten flags) rather than concentrating on one task the way the household-scale study's appeals concentrated on a single chore.

Discussion

The pattern we observe is close to a mirror image of the one we set out to replicate. Where a household's appeals process raised the concordance between a chore wheel's assignment and who completed it while leaving completion itself untouched, a working bee's public check-off board leaves concordance untouched while raising completion. One reading consistent with both findings is that a household already has a high-trust channel — direct conversation, shared consequences — through which real completion gets negotiated, so an appeals process's only remaining work is tidying the record; a building's residents mostly lack that channel, so a public board's main available lever is the audience it supplies, which several literatures treat as sufficient on its own to raise contribution to a shared task [3], [4], [10], without needing to touch the record at all. Social-loafing accounts, which would predict a weaker household-style effect at building scale rather than a different one entirely, do not obviously explain why the effect inverted rather than merely shrank [5]; we do not have a clean way to adjudicate between these accounts from a between-buildings design alone.

Limitations

Buildings were not randomised to arm; each partnering firm rolled the board out on its own schedule, and we cannot rule out that firms serving buildings already inclined toward higher working-bee turnout adopted the board earlier. Rollout order was, so far as our partnering firms could establish, unrelated to any building's prior working-bee attendance record, and the completion gap's consistency across three firms and four cycles suggests it is not an artefact of any single firm's practice. Our 42-building sample is modest next to the household study's 76 households, though a between-buildings design necessarily costs more per unit than a within-household one, and the gap's steady widening across all four cycles argues against a single seasonal or scheduling coincidence. We audited three tasks common to nearly every building's working bee; tasks requiring specialist contractors were out of scope, and we make no claim about whether a public board would move completion on tasks residents cannot themselves perform.

Conclusion

An accountability mechanism modelled closely on one that moved a household's record without its behaviour instead moved a building's behaviour without its record. We do not read this as evidence against the original finding, which concerned a different unit of analysis under different social conditions, but as evidence that the same mechanism's effect is not portable across scale without also asking what channel, if any, already exists for the outcome the mechanism is meant to secure. Future work might test whether the dispute-flag feature alone, stripped of the board's public visibility, reproduces either pattern, and whether the completion effect we report persists once a board's novelty has worn off across further cycles.

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